

SKEL 4223: Digital Signal Processing *with* MATLAB

CHAPTER 2

DISCRETE-TIME SYSTEM

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1.0 Linearity

```
n = -2:4
x1 = [0 0 1 3 0 2 0]
x2 = [1 1 2 3 2 2 1]

% System 1 test
y1a = sistem1(x1+x2)           % first test
y1b = sistem1(x1)+sistem1(x2) % second test
dispresult(y1a,y1b)

% System 2 test
y2a = sistem2(x1+x2)           % first test
y2b = sistem2(x1)+sistem2(x2) % second test
dispresult(y2a,y2b)
```

```
% System 1:  $y[n]=x[n]^2$ 
function y = sistem1(x)
    y = x.^2;
end

% System 2:  $y[n]=x[n]+x[n-1]$ 
function y = sistem2(x)
    y = [x 0] + [0 x];
end

% Displaying Result
function dispresult(ya,yb)
    if ~isequal(ya,yb)
        disp('sistem is not linear')
    else
        disp('sistem is linear')
    end
end
end
```

1.1 Introduction

A linear system obeys superposition principle as below:

For $y_a[n] = T\{x_1[n] + x_2[n]\}$ and $y_b[n] = T\{x_1[n]\} + T\{x_2[n]\}$,
 $y_a[n] = y_b[n]$ must be true.

Let's consider the following signals to demonstrate linearity of a system.

```
n = -2:4
x1 = [0 0 1 3 0 2 0]
x2 = [1 1 2 3 2 2 1]
```

n =							
	-2	-1	0	1	2	3	4
x1 =							
	0	0	1	3	0	2	0
x2 =							
	1	1	2	3	2	2	1

1.2 Testing the Superposition Principle

From previous section, there are two tests for the superposition principle. First is to sum two signals and feed the result into the system to be tested. Second test is by first feeding the two signals into the tested system separately and sum the two output signals afterward.

```
% System 1 test
y1a = sistem1(x1+x2)           % first test
y1b = sistem1(x1)+sistem1(x2)  % second test
dispresult(y1a,y1b)

% System 2 test
y2a = sistem2(x1+x2)           % first test
y2b = sistem2(x1)+sistem2(x2)  % second test
dispresult(y2a,y2b)
```

1.3 System 1: $y[n] = x[n]^2$

```
function y = sistem1(x)
    y = x.^2;
end
```

y1a =

1 1 9 36 4 16 1

y1b =

1 1 5 18 4 8 1

system is not linear

1.4 System 2: $y[n] = x[n] + x[n - 1]$

```
function y = sistem2(x)
    y = [x 0] + [0 x];
end
```

y2a =

1 2 4 9 8 6 5 1

y2b =

1 2 4 9 8 6 5 1

system is linear

1.5 Displaying the Result

```
function dispresult(ya,yb)
    if ~isequal(ya,yb)
        disp('system is not linear')
    else
        disp('system is linear')
    end
end
```

2.0 Time-Invariant

```

n1 = [-1 0 1];
x1 = [1 2 1];
n2 = [1 2 3 4 5 6 7];
x2 = [1 2 1 2 1 2 1];
nd = 2;

% Testing Signal x1 with System 1
[y1a,n1a] = sistem1(x1,n1+nd);           % Delay --> system
[y1b,n1b] = sistem1(x1,n1); n1b = n1b+nd; % system --> delay
dispresult(n1,x1,n1a,y1a,n1b,y1b)

% Testing Signal x2 with System 1
[y2a,n2a] = sistem1(x2,n2+nd);           % Delay --> system
[y2b,n2b] = sistem1(x2,n2); n2b = n2b+nd; % system --> delay
dispresult(n2,x2,n2a,y2a,n2b,y2b)

% Testing Signal x2 with System 2
[y2a,n2a] = sistem2(x2,n2+nd);           % Delay --> system
[y2b,n2b] = sistem2(x2,n2); n2b = n2b+nd; % system --> delay
dispresult(n2,x2,n2a,y2a,n2b,y2b)
    
```

```
% Testing Signal x2 with System 3
```

```
[y2a,n2a] = sistem3(x2,n2+nd);           % Delay --> system
[y2b,n2b] = sistem3(x2,n2); n2b = n2b+nd; % system --> delay
dispresult(n2,x2,n2a,y2a,n2b,y2b)
```

```
% System 1
```

```
function [y,n] = sistem1(x,n)
    pad = 10;
    x = [zeros(1,pad) x zeros(1,pad)];
    n = n(1)-pad:n(end)+pad;
    idx = 1-n(1);
    y = zeros(1,10);
    for ne = n(1):n(end)
        if ne>=0, nsum = (0:ne)+idx;
        else,     nsum = (ne:0)+idx;
        end
        y(ne+idx) = sum(x(nsum));
    end
end
```

```
% System 2
```

```
function [y,nout] = sistem2(x,n)
    pad = 4;
    x = [zeros(1,pad) x zeros(1,pad)];
    n = n(1)-pad:n(end)+pad;
    idx = 1-n(1);
    nout = n(1)+abs(rem(n(1),2)):2:n(end)-abs(rem(n(end),2));
    y = x(nout+idx);
    nout = nout/2;
end
```

```
% System 3
```

```
function [y,nout] = sistem3(x,n)
    pad = 4;
    x = [zeros(1,pad) x zeros(1,pad)];
    nout = n(1)-pad:n(end)+pad;
    N = length(x);
    y = zeros(1,N);
    for m=2:N
        y(m)=0.5*(x(m)+x(m-1));
    end
end
```

% Displaying the Results

```
function dispresult(n,x,na,ya,nb,yb)
    xmin = min([n(1) na(1) nb(1)]);
    xmax = max([n(end) na(end) nb(end)]);
    ymax = max([x ya yb]);
    subplot(3,1,1), stem(n,x)
    xlim([xmin xmax]), ylim([0 ymax])
    xlabel('Time, n'), ylabel('Amplitude')
    title('Input x[n]')
    subplot(3,1,2), stem(na,ya)
    xlim([xmin xmax]), ylim([0 ymax])
    xlabel('Time, n'), ylabel('Amplitude')
    title('Output y[n]: Time-invariant First test')
    subplot(3,1,3), stem(nb,yb)
    xlim([xmin xmax]), ylim([0 ymax])
    xlabel('Time, n'), ylabel('Amplitude')
    title('Output y[n]: Time-invariant Second test')
end
```

2.1 Introduction

Time-invariant system is a system that does not change when a specific delay is applied either to the input or output of the system where.

$$T\{x[n - n_d]\} = y[n - n_d]$$

In this example, we will use the following signal x1 and x2. The test delay is set as nd=2.

```
n1 = [-1 0 1];  
x1 = [1 2 1];  
n2 = [1 2 3 4 5 6 7];  
x2 = [1 2 1 2 1 2 1];  
nd = 2;
```

2.2 Time-Invariant Test

```

% Testing Signal x1 with System 1
[y1a,n1a] = sistem1(x1,n1+nd);           % Delay --> system
[y1b,n1b] = sistem1(x1,n1); n1b = n1b+nd; % system --> delay
dispresult(n1,x1,n1a,y1a,n1b,y1b)

% Testing Signal x2 with System 1
[y2a,n2a] = sistem1(x2,n2+nd);           % Delay --> system
[y2b,n2b] = sistem1(x2,n2); n2b = n2b+nd; % system --> delay
dispresult(n2,x2,n2a,y2a,n2b,y2b)

% Testing Signal x2 with System 2
[y2a,n2a] = sistem2(x2,n2+nd);           % Delay --> system
[y2b,n2b] = sistem2(x2,n2); n2b = n2b+nd; % system --> delay
dispresult(n2,x2,n2a,y2a,n2b,y2b)

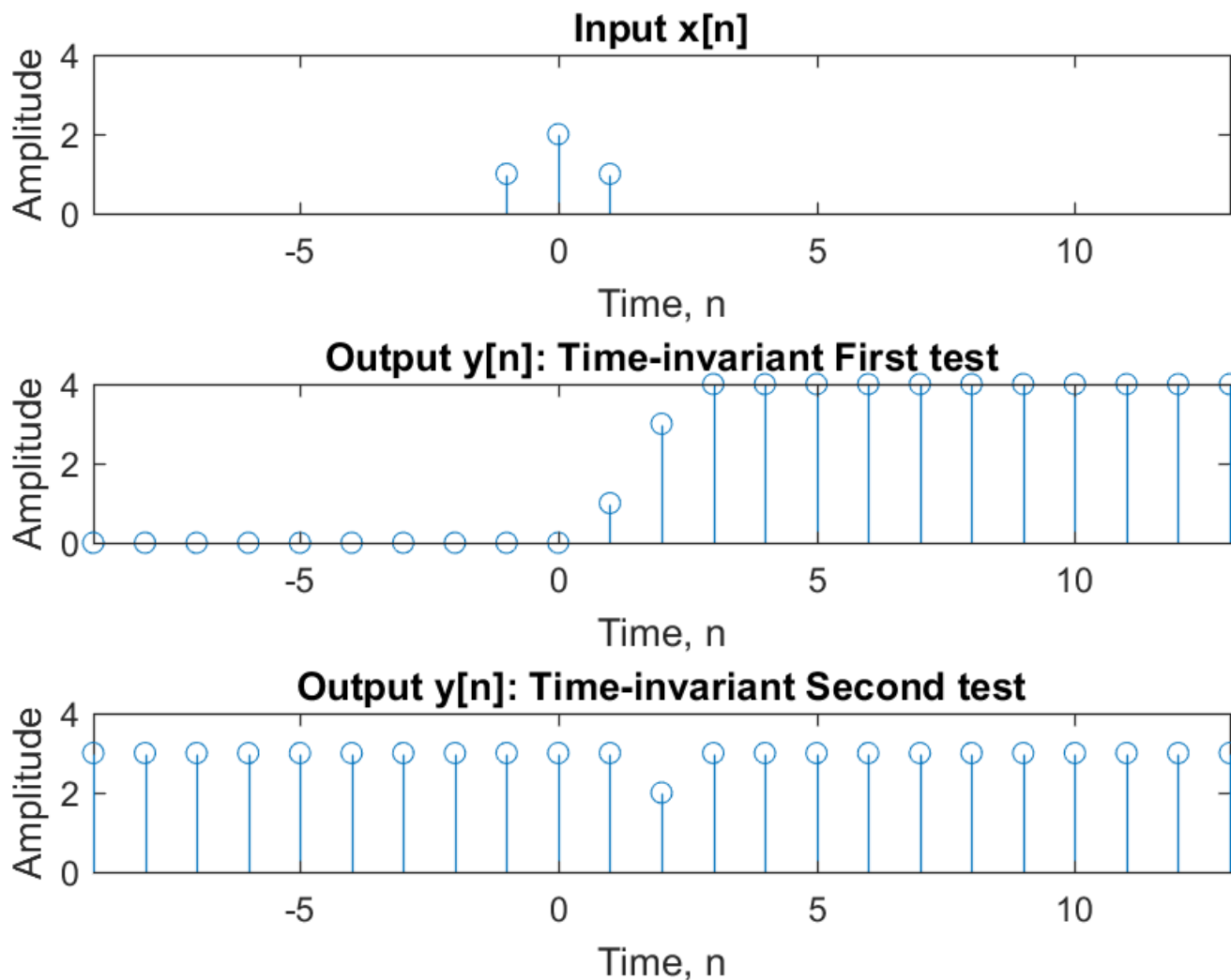
% Testing Signal x2 with System 3
[y2a,n2a] = sistem3(x2,n2+nd);           % Delay --> system
[y2b,n2b] = sistem3(x2,n2); n2b = n2b+nd; % system --> delay
dispresult(n2,x2,n2a,y2a,n2b,y2b)
    
```

2.3 System 1: $y[n] = \sum_{k=0}^n x[k]$

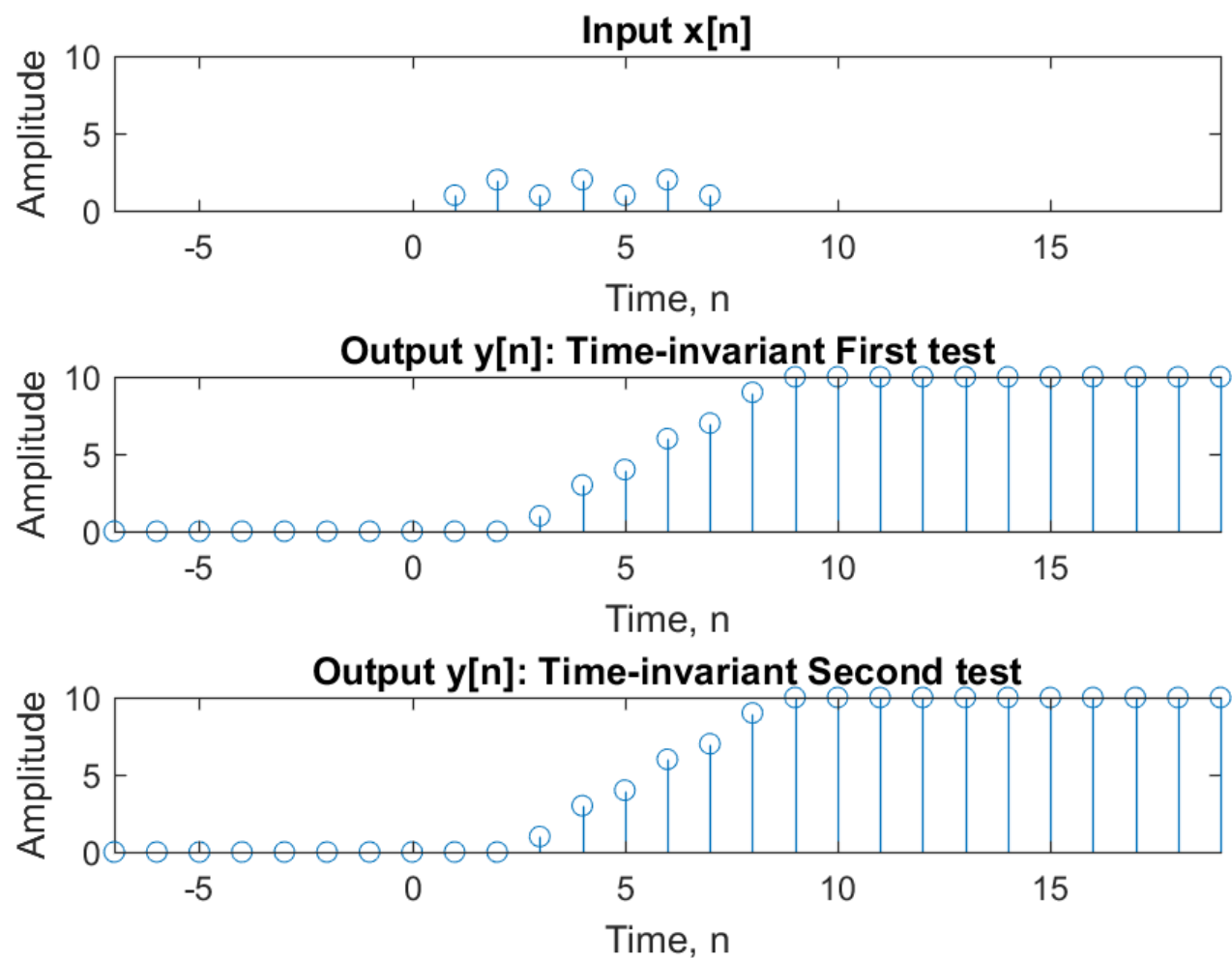
This system sum input signal $x[n]$ from $n = 0$ to the current n evaluated for $y[n]$

```
function [y,n] = sistem1(x,n)
    pad = 10;
    x = [zeros(1,pad) x zeros(1,pad)];
    n = n(1)-pad:n(end)+pad;
    idx = 1-n(1);
    y = zeros(1,10);
    for ne = n(1):n(end)
        if ne>=0, nsum = (0:ne)+idx;
        else,      nsum = (ne:0)+idx;
        end
        y(ne+idx) = sum(x(nsum));
    end
end
```

Result of testing $x_1[n]$ with System 1. The system is not time-invariant.



Even when test on x_2 shows equal output for test 1 and test 2, this system is not time-invariant since there are other signal when tested shows different output for the two tests (proven by the previous test).

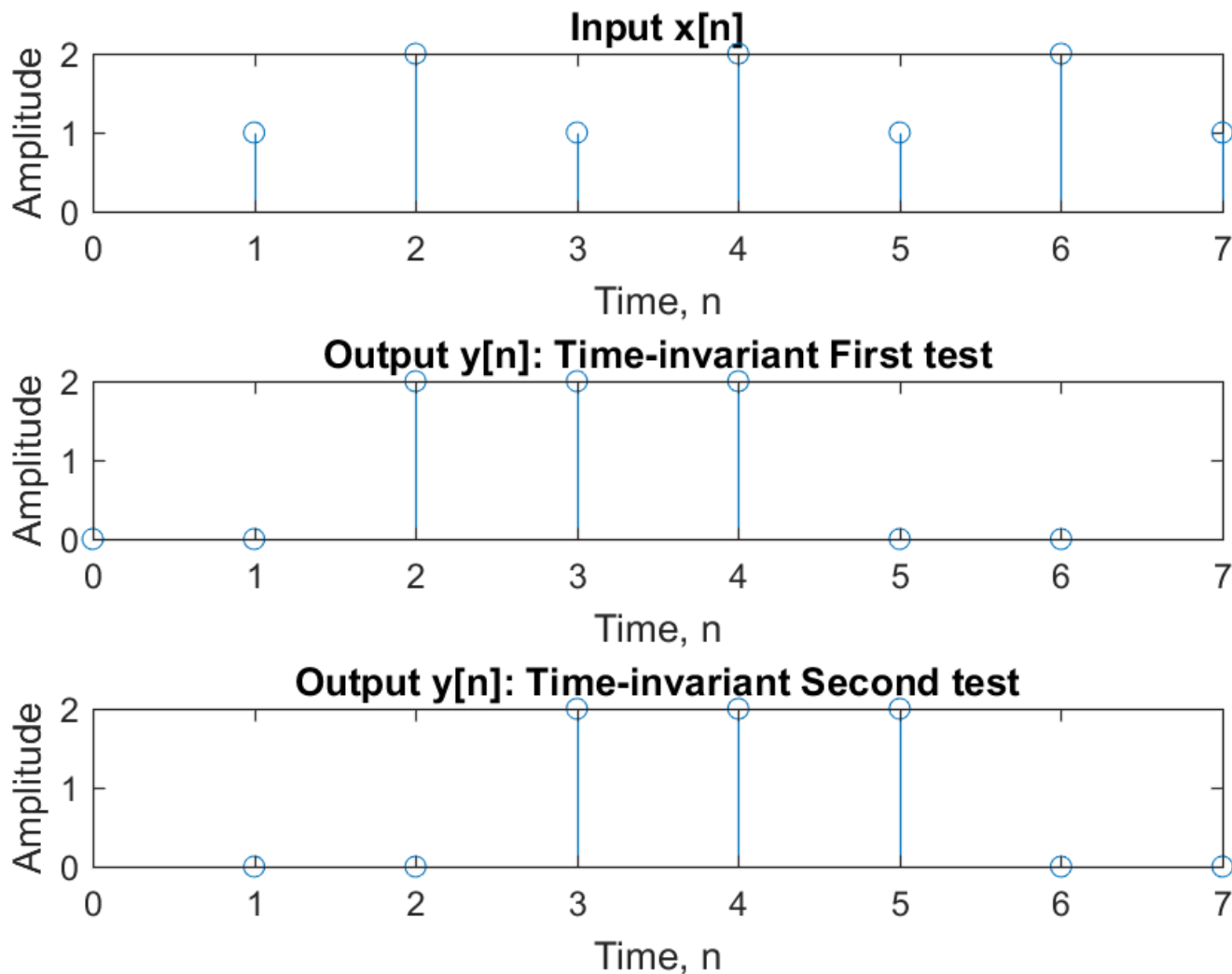


2.4 System 2: $y[n] = x[2n]$

This output $y[n]$ of the system takes all even samples of the input signal $x[n]$.

```
function [y,nout] = sistem2(x,n)
    pad = 4;
    x = [zeros(1,pad) x zeros(1,pad)];
    n = n(1)-pad:n(end)+pad;
    idx = 1-n(1);
    nout = n(1)+abs(rem(n(1),2)):2:n(end)-abs(rem(n(end),2));
    y = x(nout+idx);
    nout = nout/2;
end
```

Result of testing $x_2[n]$ with System 2. The system is not time-invariant

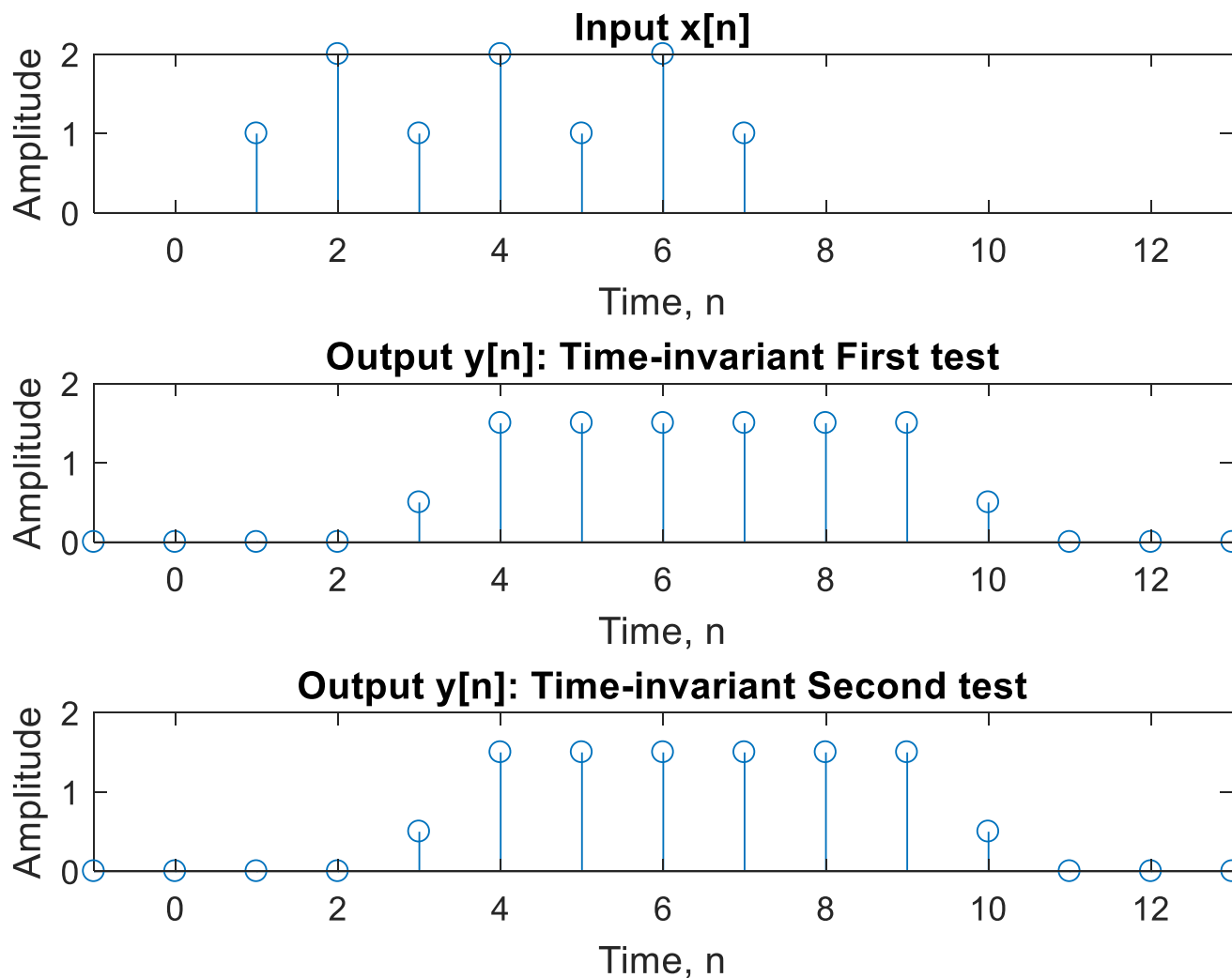


2.5 System 3: $y[n] = 0.5(x[n] + x[n - 1])$

This is an averaging system of 2 input samples of $x[n]$.

```
function [y,nout] = sistem3(x,n)
    pad = 4;
    x = [zeros(1,pad) x zeros(1,pad)];
    nout = n(1)-pad:n(end)+pad;
    N = length(x);
    y = zeros(1,N);
    for m=2:N
        y(m)=0.5*(x(m)+x(m-1));
    end
end
```

Result of testing $x_2[n]$ with System 3. The system is time-invariant



2.6 Displaying the Result

```
function dispresult(n,x,na,ya,nb,yb)
    xmin = min([n(1) na(1) nb(1)]);
    xmax = max([n(end) na(end) nb(end)]);
    ymax = max([x ya yb]);
    subplot(3,1,1), stem(n,x)
    xlim([xmin xmax]), ylim([0 ymax])
    xlabel('Time, n'), ylabel('Amplitude')
    title('Input x[n]')
    subplot(3,1,2), stem(na,ya)
    xlim([xmin xmax]), ylim([0 ymax])
    xlabel('Time, n'), ylabel('Amplitude')
    title('Output y[n]: Time-invariant First test')
    subplot(3,1,3), stem(nb,yb)
    xlim([xmin xmax]), ylim([0 ymax])
    xlabel('Time, n'), ylabel('Amplitude')
    title('Output y[n]: Time-invariant Second test')
end
```